



Review

Vector valued Functions and space curves

Vector valued Functions

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

Vector (-valued) Function: domain $\rightarrow$ range
 $\nwarrow$ set of real numbers $\nwarrow$ set of vectors

$$t \mapsto \vec{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k} \\ = \langle f(t), g(t), h(t) \rangle$$

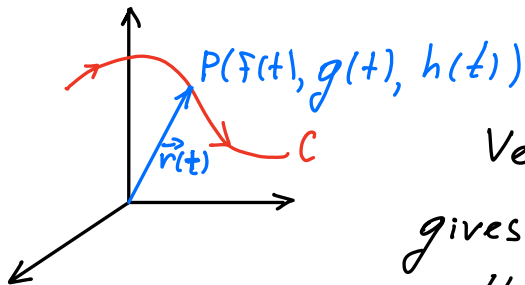
Rmk: domain = values t for which $\vec{r}(t)$ is defined.

Space curves

Set of points (x, y, z) with $t \in I$ interval - "space curve" C .

given functions $\swarrow$
 $x = f(t), y = g(t), z = h(t)$

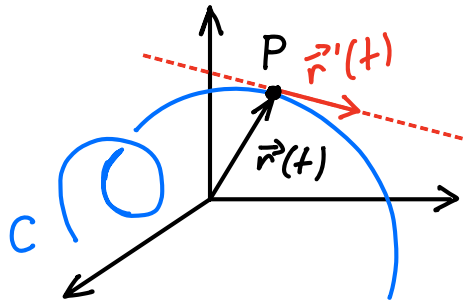
$\swarrow$ parametric equations of C ,
 t -parameter



Vector function $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ gives a "moving" vector, whose tip traces the curve C .

Calculus of space curves

Derivatives: $\vec{r}(t)$ vector function $\leadsto \frac{d\vec{r}}{dt} = \vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$



- $\vec{r}'(t)$ - tangent vector to curve C at point P (if $\vec{r}'(t) \neq 0$ and exists)
- tangent line to C at P - line through P , parallel to $\vec{r}'(t)$

Integrals:

$$\int_a^b \vec{r}(t) dt = \left(\int_a^b f(t) dt \right) \vec{i} + \left(\int_a^b g(t) dt \right) \vec{j} + \left(\int_a^b h(t) dt \right) \vec{k}$$

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

Fund. Thm of Calculus
(in vector setting)

$$\int_a^b \vec{r}(t) dt = \vec{R}(t) \Big|_a^b = \vec{R}(b) - \vec{R}(a)$$

$$\nwarrow \vec{R}'(t) = \vec{r}(t)$$

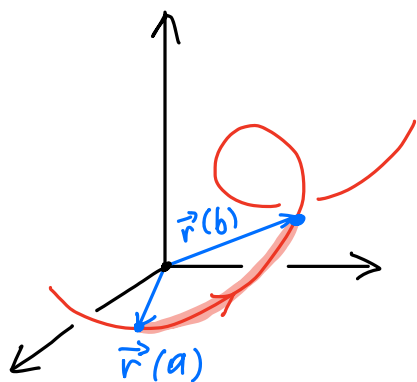
We denote $\vec{R}(t) = \int \vec{r}(t) dt$ (indefinite integral)

Arc length & TNB Frame

$$L = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt$$

length of the curve

$$= \int_a^b |\vec{r}'(t)| dt$$



$$C: \vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

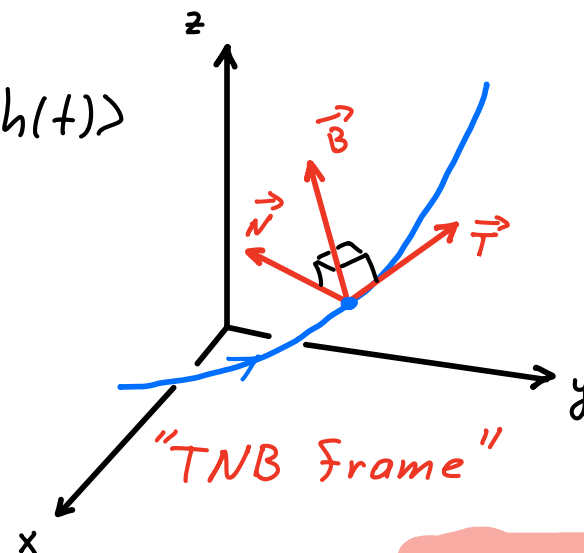
unit

$$\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}$$

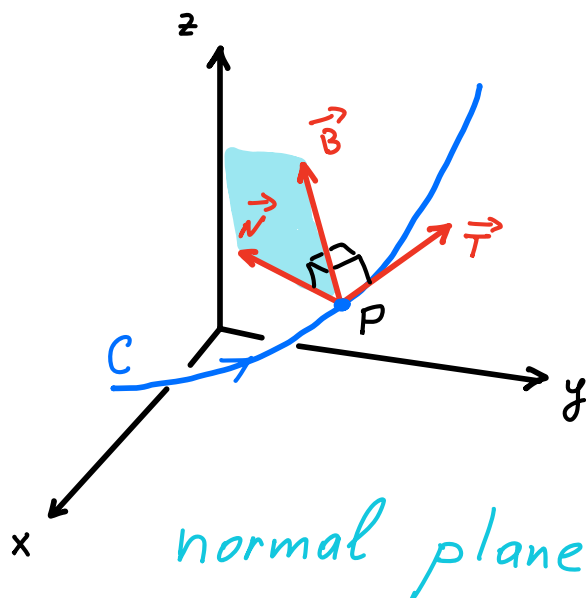
normal

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$$

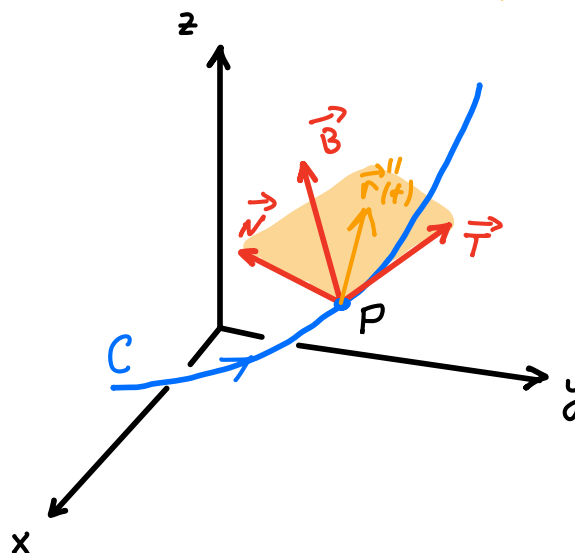
binormal



Rmk: $T \times N = B$
 $T = N \times B$
 $N = B \times T$

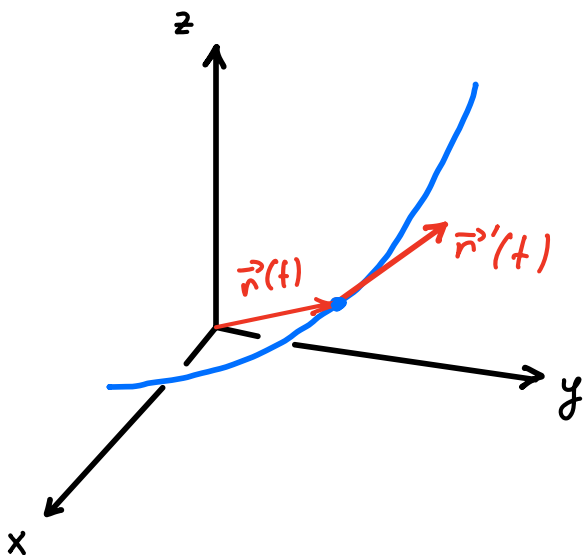


osculating plane



Rmk: $\vec{r}''(t)$ is in the osculating plane.

Motion in space



particle moving through space;
position at time t : $\vec{r}(t)$

$$\vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h} = \vec{v}(t)$$

displacement vector
per unit time $\uparrow$ velocity vector

speed: $|\vec{v}(t)| = |\vec{r}'(t)| = \frac{ds}{dt} \leftarrow$ arc length fct.

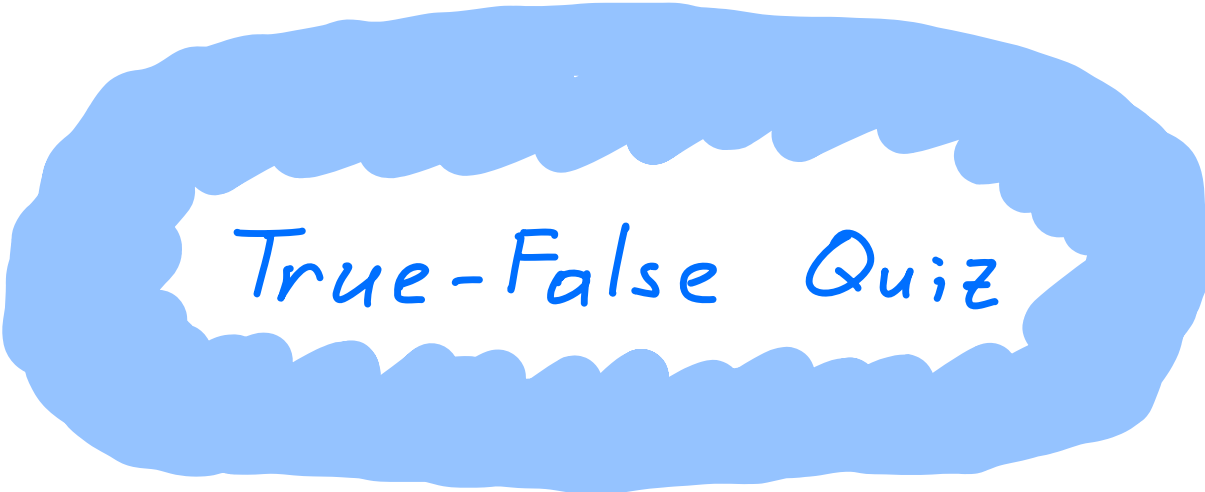
$\uparrow$ rate of change of dist. w.r.t. time

Reminder: $s(t) = \int_a^t |r'(u)| du \Rightarrow \frac{ds}{dt} = |r'(t)|$

acceleration: $\vec{a}(t) = \vec{v}'(t) = \vec{r}''(t)$

Rmk: If we know $\vec{a}(t)$ and $\vec{r}(t_0), \vec{v}(t_0)$ (initial position and velocity) then

$$\vec{v}(t) = \int_{t_0}^t \vec{a}(u) du + \vec{v}(t_0), \quad \vec{r}(t) = \int_{t_0}^t \vec{v}(u) du + \vec{r}(t_0)$$



True-False Quiz

True-False Quiz: 1. Let $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, then

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle.$$

2. $\vec{r}(t) = t^3 \vec{i} + 2t^3 \vec{j} + 3t^3 \vec{k}$ is a line.

3. The curve $\vec{r}(t) = \langle 2t, 3-t, 0 \rangle$ is a line
passes through the origin.

4. $(\vec{u}(t) \times \vec{v}(t))' = \vec{u}'(t) \times \vec{v}'(t)$

5. The binormal vector $\vec{B}(t) = \vec{N}(t) \times \vec{T}(t)$

6. If $|\vec{r}(t)| = 1$ for all t , then $\vec{r}'(t)$ is orthogonal
to $\vec{r}(t)$ for all t

7. If $|\vec{r}(t)| = 1$ for all t , then $|\vec{r}'(t)|$ is a constant

8. $\vec{r}''(t) = \frac{\vec{r}'(t) \cdot \vec{r}''(t)}{|\vec{r}'(t)|} \vec{T}(t) + \frac{\vec{r}'(t) \times \vec{r}''(t)}{|\vec{r}'(t)|} \vec{N}(t)$



Few more problems

Problems:

- ① Describe by a vector equation the intersection C
of $z = x^2 + y^2$ -paraboloid
and $y = x^2$ -parabolic cylinder
- ② Find equations of normal and osculating planes
for $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j} + t \vec{k}$ at $P(0, 1, \frac{\pi}{2})$
- ③ $\vec{r}(t) = \langle t^2, \frac{2}{3}t^3, t \rangle$. Find $\vec{T}, \vec{N}, \vec{B}$ at $P(1, \frac{2}{3}, 1)$.